brown open silicon

designing open chips. together.

agenda

welcome back & icebreaker

club structure & setup

chip design flow

entry level projects (xinting)

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welcome back & icebreaker

If you had to live in one Brown building for the rest of your life (assuming food is included), which one would it be?

name + year + concentration + interests

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the only right answer

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One thing I know
One thing I want to learn
One thing I can help with

what do you want to accomplish?

club structure & setup

most likely fridays/saturdays (if people can make it)

schedule is being finalized (mixture of work time, crash courses, Q&A, industry chats)

fill out this when2meet with availability!

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club structure & setup

google drive for resources, tutorials, slides
google chat for discussions, questions, etc

potentially other forms of communication (up to us!)

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club structure & setup

support needed:
community outreach lead
website development

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club structure & setup

project structure

attendees (1 hour per week)

sit in & learn about chip design
listen to industry leaders

project work groups (time varies)

sit in & learn about chip design
listen to industry leaders
working on developing a chip

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club structure & setup

project work groups

teams of 2–4

tool onboarding
monthly design checkups/project support
debugging help
potential research publications & travel grants

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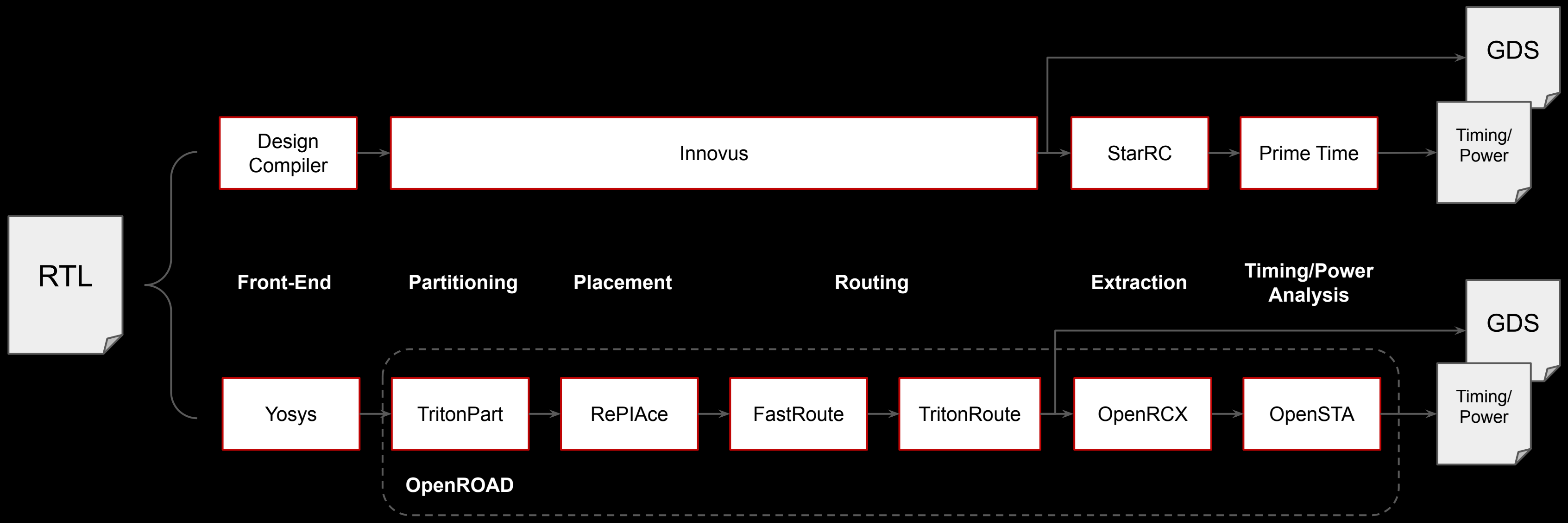
chip design flow

Digital flow:



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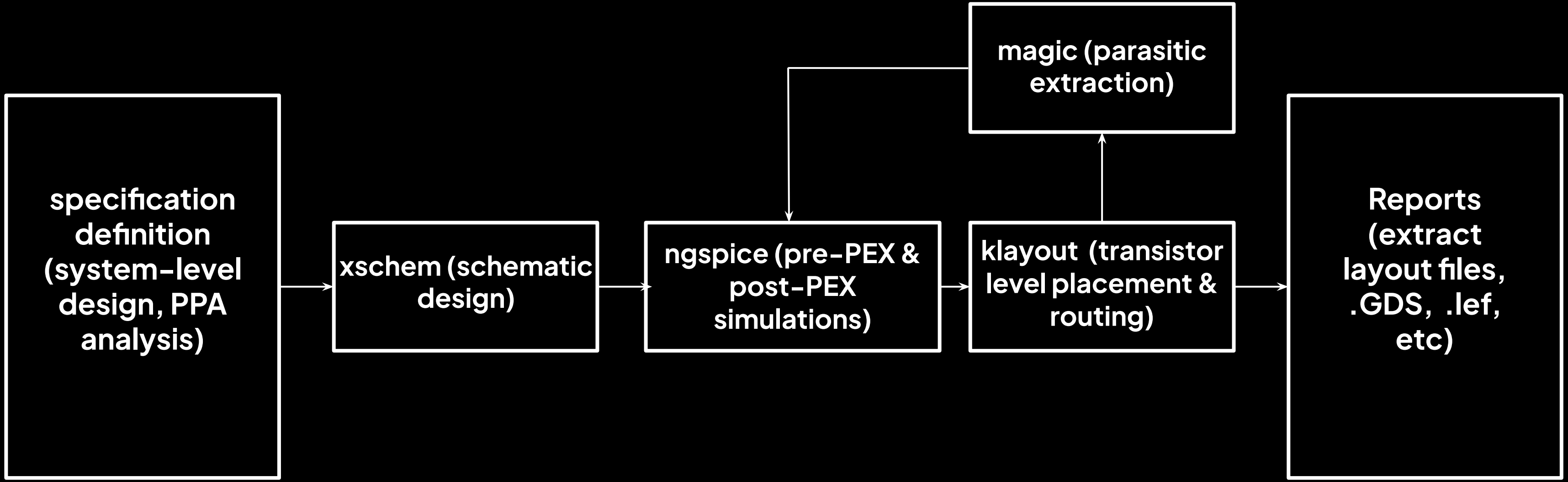
chip design flow

Custom flow:



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brainstorming tips

Think outside of the box
Define the design goals clearly
Most viable product(MVP) first
Make it modular
Connect with people

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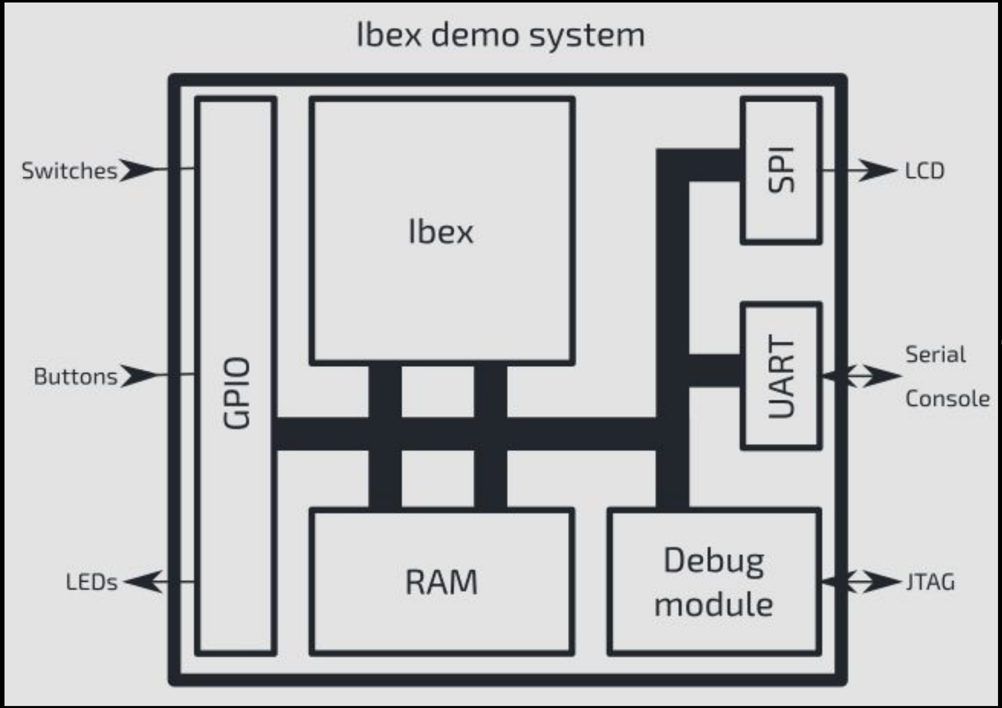
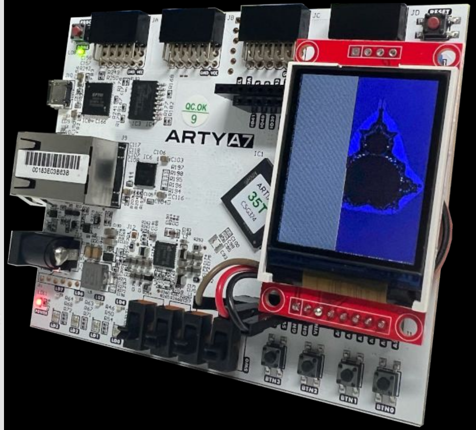
entry level projects

Digital design: RISC-V SoC

[Ibex demo system]





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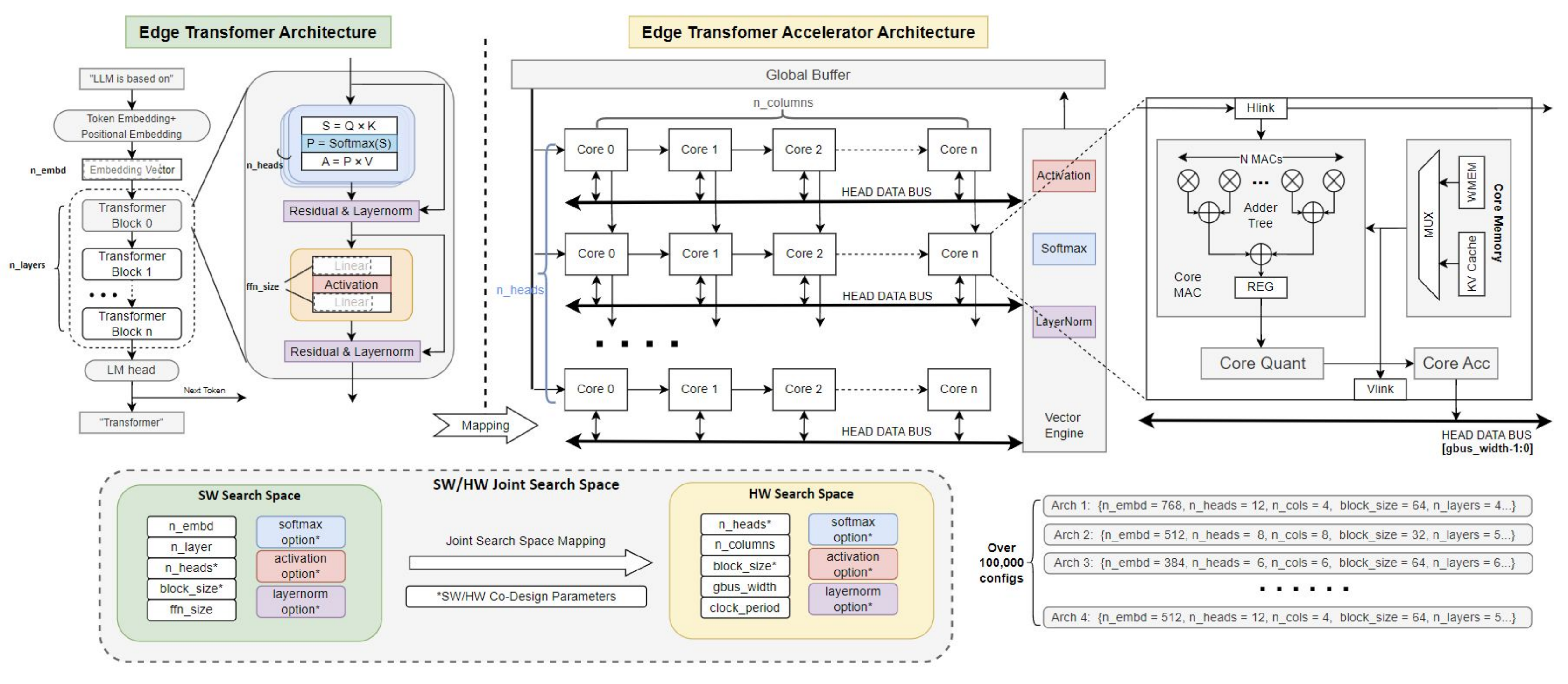
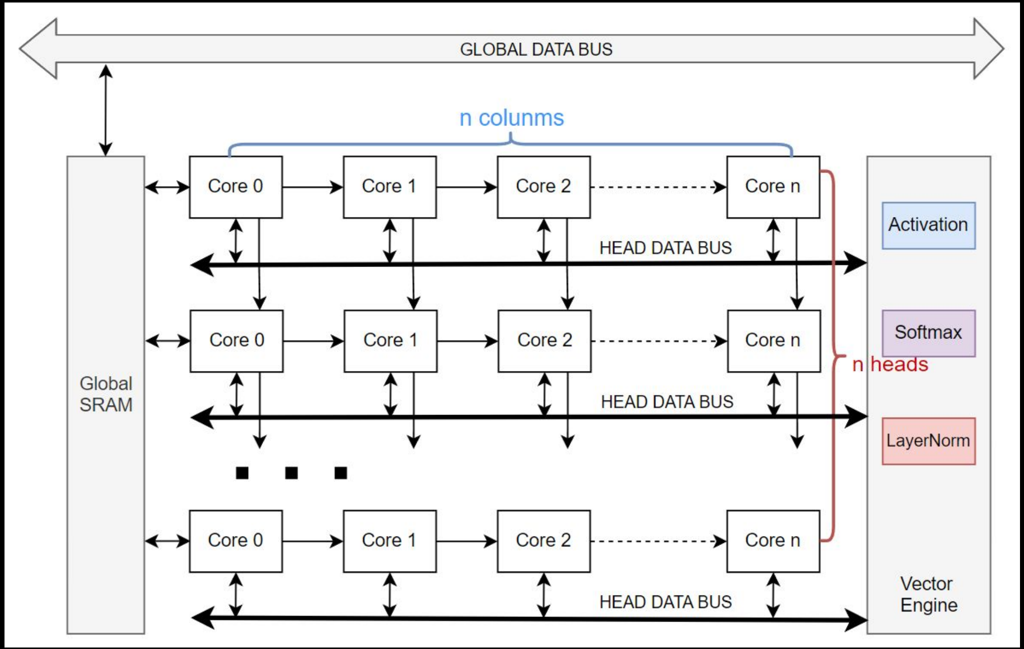
entry level projects

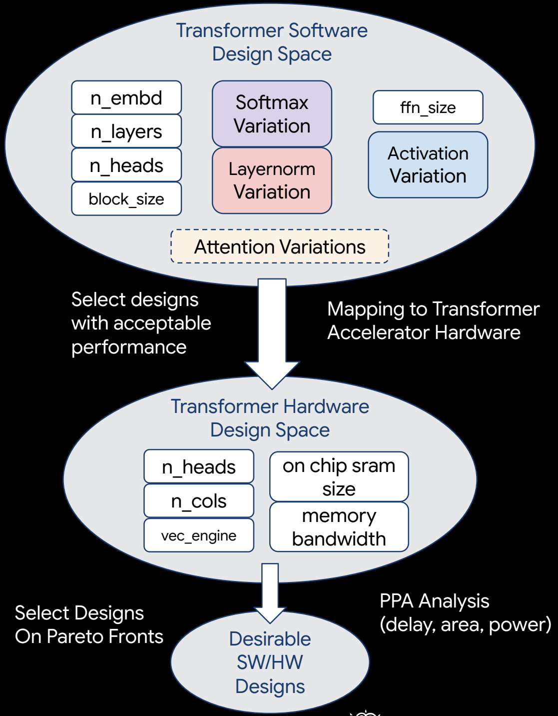
Digital design: systolic array



entry level projects

Digital design: design space exploration





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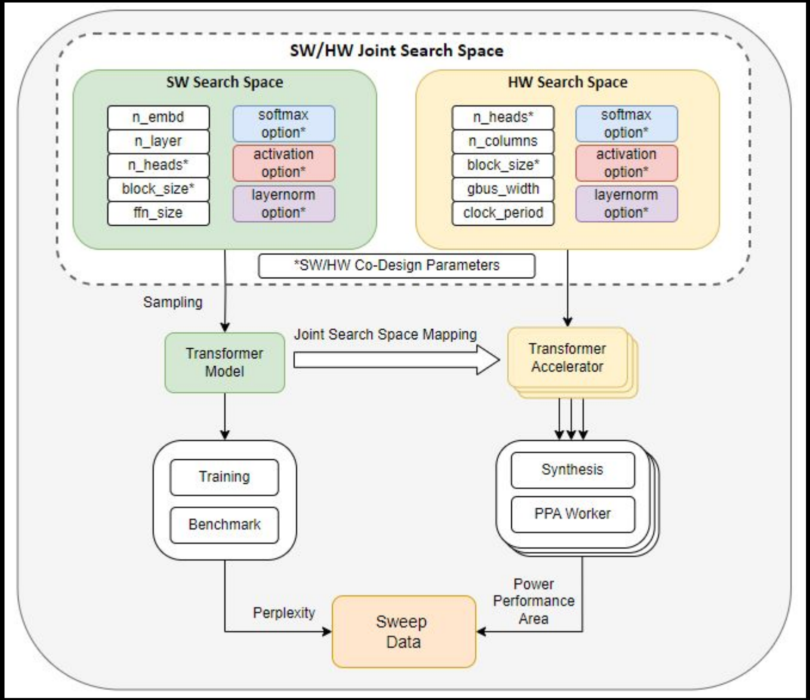
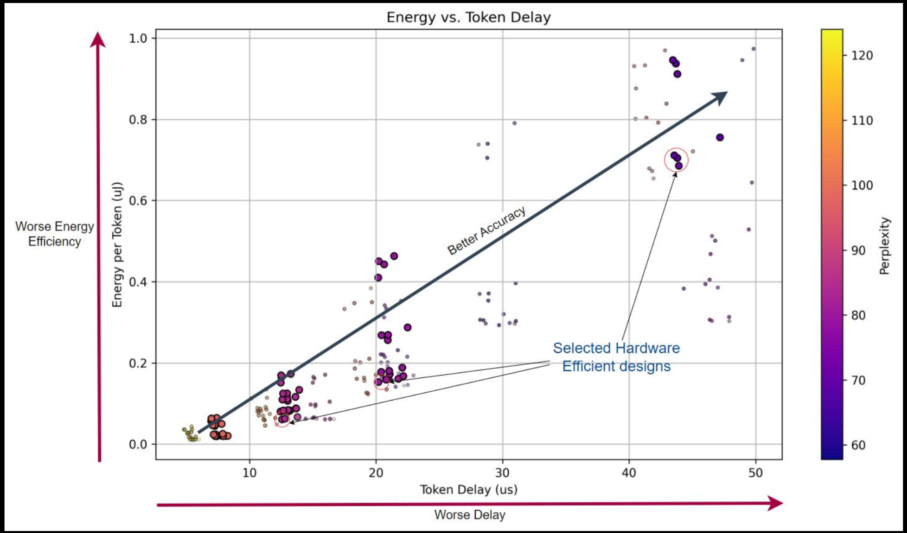
entry level projects

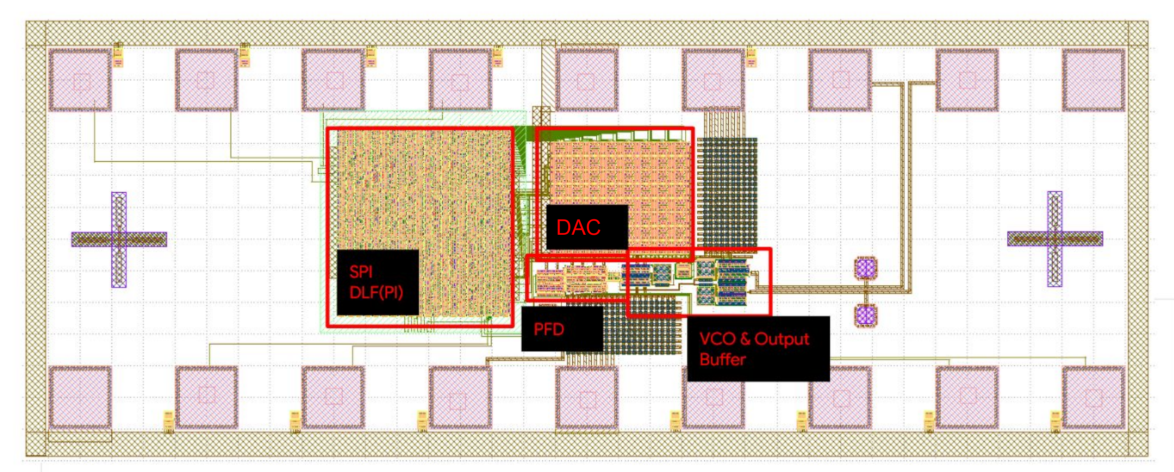
Digital design: design space exploration

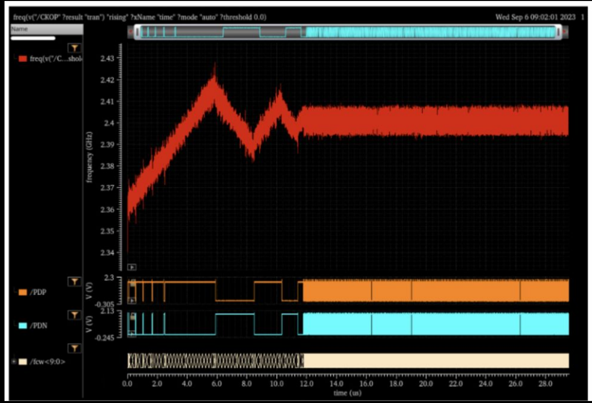




entry level projects

Custom design: PLL(Phased-locked Loop)

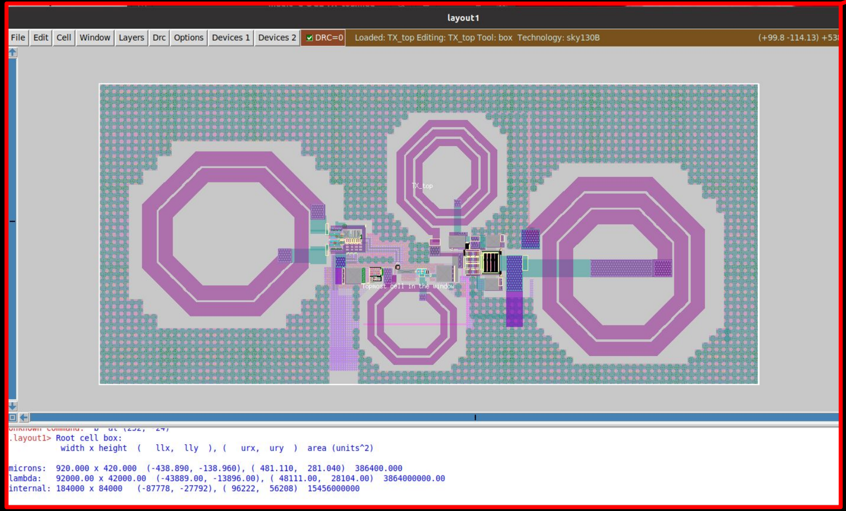




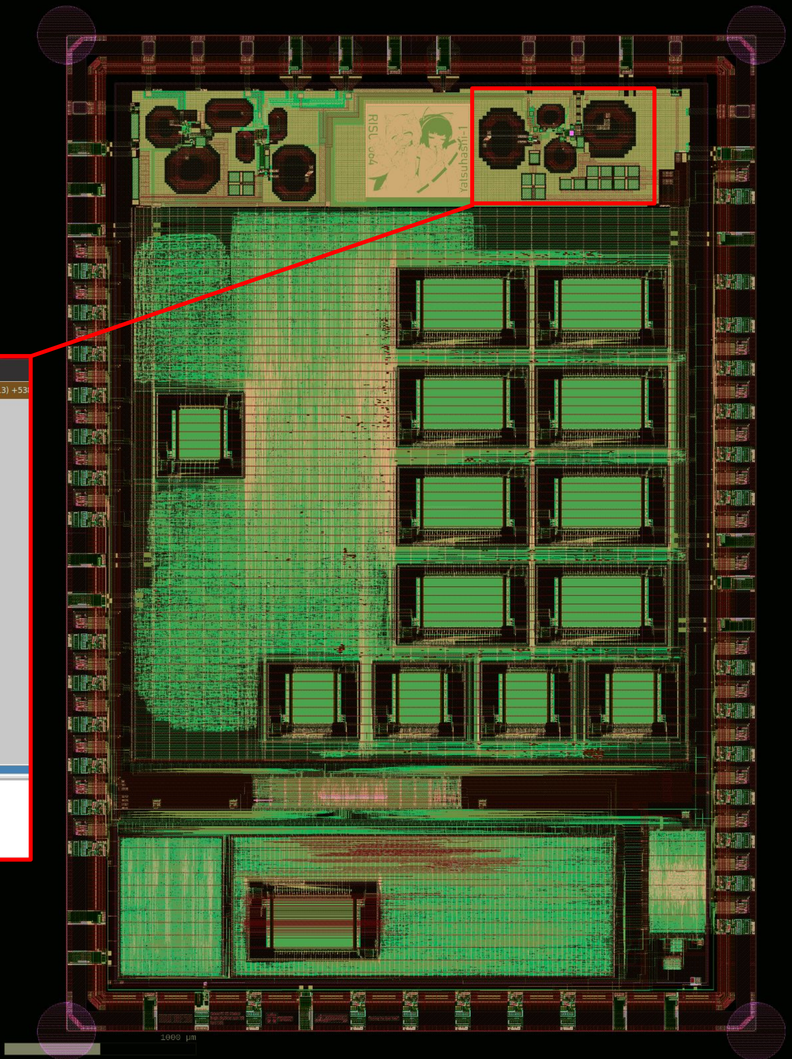
Input : 240MHz Signal
Output: 2.4GHz Signal

Advanced projects

RF Design: 5GHz TRX



Stacked Class-E Power Amplifier



5GHz
TRX

Dual-Issue
Risc-V

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thank you for coming!

choose your project interest (non-binding)

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